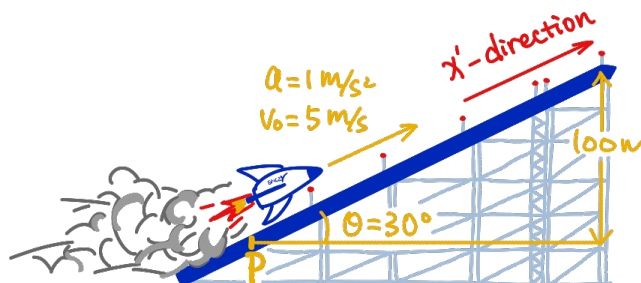


Physics 5A Discussion (Week 2)

Department of Physics and Astronomy, UCLA



Problem 1: Jinyan is launching a rocket using a ramp at an angle of $\theta = 30^\circ$ with respect to the horizontal. When the rocket reaches point P as shown, it has an instantaneous velocity $v_0 = 5 \text{ m/s}$ in the x' -direction, and its acceleration in the x' -direction is kept constant at 1 m/s^2 . How long does it take for the rocket to reach a vertical displacement of $h = 100 \text{ m}$ (measured with respect to the height of point P)?

This is quite a complicated problem, but we can break it down into a few steps:

Step 1: Looking at the triangle shown, how far does the rocket need to travel in the x' -direction (on the ramp) to reach $h = 100 \text{ m}$?

Step 2: Draw the acceleration vs. time graph, with the acceleration in the x' -direction on the vertical axis, and time on the horizontal axis. (*Hint: As mentioned in the problem, acceleration is kept constant over time.*)

Step 3: From step 2, draw the velocity vs. time graph for the first 20 seconds, and write down the mathematical expression for the instantaneous velocity as a function of time. (*Hint: The change in velocity is the “area” under the acceleration curve.*)

Step 4: From step 3, deduce the shape of curve in the position vs. time graph. Is it a straight line, parabola, or an exponential curve? (*Hint: The slope of your position curve at each time instant should be the value of your velocity curve at the same time instant.*)

Step 5: Again using your velocity vs. time graph, how far has the rocket traveled in the x' -direction in the first 20 seconds? (*Hint: Use the "area" under the velocity curve to find the displacement.*)

Step 6: Recall the distance calculated in step 1, how long does the rocket take to travel such a distance? (*Hint: Try an arbitrary final time t_f and repeat from step 3. You will need the quadratic formula when computing your final answer.*)

Step 7: Now you have seen that *position curve looks like a parabola if we have a constant acceleration over time*, this result leads to the kinematic equation that you will see very soon during lecture:

$$x = \frac{1}{2}at^2 + v_0t + x_0 ,$$

where x_0 is your initial position at time $t = 0$ second, and x is your final position at a generic time t . Using this formula, find the time t for the rocket to travel from $x_0 = 0$ m to $x =$ (your result in step 1) in the x' -direction. (*Hint: You will need the quadratic formula. The result you get here should agree with the one you found in step 6.*)

Problem 2: For the rocket in problem 1, how far has the rocket traveled in the horizontal direction (x -direction) when it reaches $h = 100$ m?